Aggressive Surgical Management of Congenital Diaphragmatic Hernia: Worth the Effort?

A Multicenter, Prospective, Cohort Study

Matthew T. Harting, MD, MS,[*†] *Laura Hollinger, MD,*[*] *Kuojen Tsao, MD,*[*†] *Luke R. Putnam, MD, MS,*[*†] *Jay M. Wilson, MD,*[‡] *Ronald B. Hirschl, MD,*[§] *Erik D. Skarsgard, MD,*[¶] *Dick Tibboel, MD, PhD,*[||] *Mary E. Brindle, MD, MPH,*[**] *Pamela A. Lally, MD,*[*†] *Charles C. Miller, PhD,*[††] *and Kevin P. Lally, MD, MS*[*†], *for the Congenital Diaphragmatic Hernia Study Group*

Objective: The objectives of this study were (i) to evaluate infants with congenital diaphragmatic hernia (CDH) that do not undergo repair, (ii) to identify nonrepair rate by institution, and (iii) to compare institutional outcomes based on nonrepair rate.

Background: Approximately 20% of infants with CDH go unrepaired and the threshold to offer surgical repair is variable.

Methods: Data were abstracted from a multicenter, prospectively collected database. Standard clinical variables, including repair (or nonrepair), and outcome were analyzed. Institutions were grouped based on volume and rate of nonrepair. Preoperative mortality predictors were identified using logistic regression, expected mortality for each center was calculated, and observed / expected (O/E) ratios were computed for center groups and compared by Kruskal-Wallis ANOVA.

Results: A total of 3965 infants with CDH were identified and 691 infants (17.5%) were not repaired. Nonrepaired patients had lower Apgar scores ($P < 0.05$) and increased incidence of anomalies ($P < 0.0001$). Low-volume centers ("Lo", n=44 total, < 10 CDH pts/yr) and high-volume centers ("Hi", n = 21) had median nonrepair rates of 19.8% (range 0%–66.7%) and 16.7% (5.1%–38.5%), respectively. High-volume centers were further dichotomized by rate of nonrepair (HiLo = 5.1–16.7% and HiHi = 17.6–38.5%), leaving 3 groups: HiLo, HiHi, and Lo. Predictors of mortality were lower birth weight, lower Apgar scores, prenatal diagnosis, and presence of congenital anomalies. O/E ratios for mortality in the HiLo, HiHi, and Lo groups were 0.81, 0.94, and 1.21, respectively ($P < 0.0001$). For every 100 CDH patients, HiLo centers have 2.73 (2.4–3.1, 95% confidence interval) survivors beyond expectation.

Conclusions: There are significant differences between repaired and nonrepaired CDH infants and significant center variation in rate of nonrepair exists. Aggressive surgical management, leading to a low rate of nonrepair, is associated with improved risk-adjusted mortality.

Keywords: congenital diaphragmatic hernia repair, congenital diaphragmatic hernia study group, center volume, congenital diaphragmatic hernia, diaphragm repair, outcomes, registry

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The management of infants with congenital diaphragmatic hernia (CDH) has improved as a result of specific advances including pressure-limited ventilation, standardized postnatal management, extracorporeal membrane oxygenation (ECMO) support, and neonatal critical care expertise. Although the overall survival rate among patients with CDH remains approximately 70%, depending upon numerous factors, select groups have suggested that further improvements in survival may be achievable through evidence-based, protocol-driven standardization of management.[1–3] Despite these encouraging results, conflicting data exist,[4] and innumerable elements, both known and unknown, influence these outcomes.[5]

Approximately 20% of infants with CDH identified in the pre or postnatal period never undergo surgical repair.[6] Surgical ineligibility is subjective, traditionally determined by persistent physiologic instability, associated congenital anomalies, anatomic limitation, or other physician-driven perceptions. This patient population remains enigmatic and infrequently studied.

A recent Canadian Pediatric Surgery Network (CAPSNet) study identified a small cohort of patients who were deemed ineligible for operative repair; however, these patients could not be discriminated from operative candidates based on demographic or physiologic criteria.[7] These data highlight that one critical determinant of patient survival—threshold for operative repair—may be controlled by patient-independent factors such as individual surgeon and/or institutional practice patterns, which vary significantly.

From the *Department of Pediatric Surgery, University of Texas McGovern Medical School and Children's Memorial Hermann Hospital, Houston, TX; †Center for Surgical Trials and Evidence-based Practice, University of Texas McGovern Medical School, Houston, TX; ‡Department of Surgery, Children's Hospital Boston, Boston, MA; §Department of Surgery, Division of Pediatric Surgery, University of Michigan and CS Mott Children's Hospital, Ann Arbor, MI; ¶Department of Surgery, Division of Pediatric Surgery, University of British Columbia, Vancouver, British Columbia, Canada; ||Department of Pediatric Surgery, Erasmus Medical Center-Sophia Children's Hospital, Rotterdam, The Netherlands; **Department of Surgery, University of Calgary, Calgary, Alberta, Canada; and ††Department of Cardiothoracic and Vascular Surgery, University of Texas McGovern Medical School, Houston, TX.

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Reprints: Matthew T. Harting, MD, MS, Department of Pediatric Surgery, University of Texas McGovern Medical School and Children's Memorial Hermann Hospital, 6431 Fannin St., MSB 5.233, Houston, TX 77030. E-mail: matthew.t.harting@uth.tmc.edu.

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We sought to utilize the Congenital Diaphragmatic Hernia Study Group (CDHSG) registry to better characterize patients who did not undergo surgical repair and compare institutional outcomes based on diaphragmatic nonrepair rate. The objectives of this study were to: (i) evaluate infants with CDH that did not undergo diaphragmatic repair and compare them with those that underwent repair, (ii) identify nonrepair rate by institution, and (iii) compare institutional outcomes based on rate of nonrepair.

METHODS

Registry Data

Data were abstracted from the CDHSG registry. The CDHSG was founded in 1995 to devise and investigate key clinical questions related to CDH with a prospective, multicenter approach.[8,9] It is a voluntary group, and all live born infants with CDH from participating institutions are entered into the registry. Data including demographics, treatments, and outcomes are collected. All data are validated and missing or aberrant data addressed through direct center contact. Periodic evaluations of data collected and critical consideration of relevant clinical questions have resulted in successive revisions of the data collection form (currently version 4). The data in this registry are provided from ~70 centers around the world and include information on multiple aspects of the clinical course of patients with CDH (appendix). The cases included in our analysis were those infants diagnosed with CDH over a 10-year period (January, 2004 to December, 2013). The CDHSG is approved by the University of Texas at Houston Center for the Protection of Human Subjects/Institutional Review Board (#HSC-MS-03–223; Ref #118886).

Data points collected were demographic data including inborn/outborn status, estimated gestational age (EGA), sex, birthweight, and prenatal diagnosis status. Patient characteristics also included Apgar scores at 1 and 5 minutes and the presence of cardiac, chromosomal, or other anomalies. Center volume was calculated as cases per year and rate of nonrepair was identified based on not performing an operative CDH repair.

The registry predefined reasons for nonrepair; however, estimating survivability and the decision for not repairing the infant were locally determined. Center nonrepair rates included both inborn and outborn infants. Categories of reasons for nonrepair included (i) unable to stabilize the patient, (ii) Patient felt to be nonsurvivable/not a candidate for ECMO, (iii) Patient felt to be survivable/not a candidate for ECMO, (iv) Patient felt to be survivable/placed on ECMO but no repair done, (v) Patient came off ECMO but was unable to be repaired. Further explanations for nonrepair included in the data set were (i) anomaly, (ii) hypoxia, (iii) hypercarbia, (iv) intraventricular hemorrhage/cerebral hemorrhage, (v) multisystem organ failure, (vi) prematurity, (vii) parental request for no further therapy, (viii) sepsis, or (ix) unable to wean from ECMO.

Statistical Analysis

To identify differences between repaired and nonrepaired infants, we compared patient demographics and baseline characteristics with Pearson χ^2 analysis, Student *t* test, or univariate analysis of variance for categorical, continuous, or multiple variables, respectively.

For the assessments of center volume effects, we framed our analytical process in terms of potential opportunity cost of repair strategy (high or low repair category) adjusted for patient mix. To do this, we used a cross-validation ‘‘learning set/test set’’ analytical framework, in which we utilized a computer-generated random number process to divide the total data set into 2 equally sized groups. For one group, the ‘‘learning set,’’ we fit a logistic regression model for mortality, in which preoperative clinical variables (EGA, birthweight, Apgar scores, prenatal diagnosis, and inborn/outborn, cardiac/chromosomal anomalies) were used to identify risk factors for mortality with the patient as the unit of analysis. This served as the model generator set. For the second group, the ‘‘test set,’’ we applied the coefficients from the logistic model to the patient data, computed a probability of mortality for each patient, and aggregated these probabilities into expected numbers of deaths for each clinical center. We then divided the observed number of deaths by the expected number of deaths in each center to arrive at an observed/expected (O/E) ratio for each center. We used a Kruskal-Wallis nonparametric analysis of variance model to aggregate and compare the O/E ratios across the 3 categories of volume-and-repair status [low volume (Lo), high volume + high rate of nonrepair (HiHi), high volume + low rate of nonrepair (HiLo)]. We used the total case volumes per center as a weighting factor for the ANOVA to return the sample size to the full cohort (not just the test set). Pairwise comparisons between the groups used Tukey's method for alpha error control. Center was the unit of analysis for the O/E ANOVA. Excess mortality was computed using the same O/E ratios, by center, normalized to center population size and categorized based on center volume and rate of nonrepair.

All data were analyzed using Stata/IC version 13.1 (Stata Corp LP, College Station, TX) or SAS version 9.4 (SAS Institute, Inc., Cary, NC).

RESULTS

Patient Characteristics

Over a 10-year period, 3959 infants with CDH were identified, and of these, 691 (17.5%) did not undergo repair. Patient baseline characteristics were significantly different between repaired and nonrepaired infants (Table 1). Patients who did not undergo repair were more likely to have lower Apgar scores at 1 and 5 minutes, have major cardiac and chromosomal anomalies, were more likely to be prenatally diagnosed, and were less likely to be outborn.

The general reason for nonrepair was indicated in 74.2% of patients. The majority of patients who were not repaired were deemed nonsurvivable or unstable (67.7%). The justifications for the choice not to repair included ‘‘nonsurvivable and not a candidate for ECMO’’ (35.5%), ‘‘unable to stabilize’’ (32.2%), ‘‘potentially survivable with ECMO, but not a candidate for ECMO’’ (20.1%), ‘‘weaned off ECMO but not repaired’’ (9.9%), and ‘‘survivable and placed on ECMO’’ but died during support (2.3%) (Fig. 1).

Approximately half (52%) of nonrepair patients had complete data stating the specific reason for nonrepair. The most common reason for nonrepair was an associated anomaly (27.3%) (Fig. 2). Intraventricular/cerebral hemorrhage (14.2%), persistent hypoxia

TABLE 1. Patient Characteristics: Repaired Versus Nonrepaired

Repaired	Yes (n = 3274)	No (n = 691)	P^*
EGA (weeks)	37.8 ± 2.1	36.4 ± 3.2	0.595
Birthweight (g)	3 ± 0.6	2.6 ± 0.7	0.752
Apgar 1 minute	5.1 ± 1.7	3.4 ± 2.2	0.006
Apgar 5 minutes	7 ± 1.9	5.1 ± 2.4	0.027
Major cardiac anomaly	6.0%	17.3%	<0.0001
Chromosomal anomaly	3.4%	16.4%	<0.0001
Prenatally Diagnosed	63.0%	79.4%	<0.0001
Outborn	58.0%	43.7%	<0.0001

*t test or χ^2.

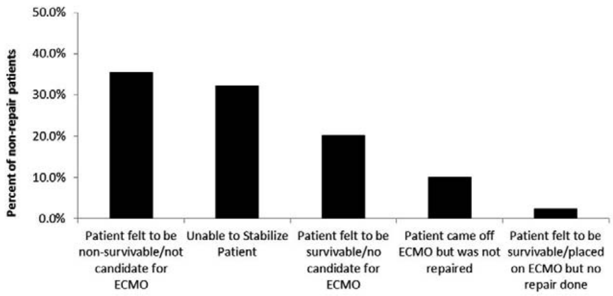

FIGURE 1. General reasons for nonrepair among all centers.

(12.3%), and parental request (10%) were also frequent reasons given by the centers for nonrepair. Seventeen percent of nonrepair infants had ''other'' listed as their reason for nonrepair and there was great variability among the explanations provided for withholding surgical therapy. For example, 2 patients were taken to surgery and deemed inoperable caused by pulmonary aplasia/diaphragmatic agenesis, and 3 patients died during preoperative planning after unanticipated changes in clinical condition.

Nonrepair Rate Across CDHSG Centers

Sixty-five centers were included in the analysis. There were 44 low volume centers (<10 cases annually, mean: 5.2 cases/year, range: 1.2–9.6 cases/year; 1729 total patients) and 21 high volume centers (≥10 cases annually, mean: 14.2 cases/year, range: 10.8–26.0 cases/year; 2230 total patients). The mean rate of nonrepair among low-volume centers was 21.5% (median 19.8%), ranging from 0% to 66.7% and the mean rate of nonrepair in high volume centers was 17.6% (median 16.7%), ranging from 5.1% to 38.5% (Fig. 3). Further subdividing the high-volume centers, the mean rate of nonrepair in the HiHi group was 23.7% (median 21.4%) and the mean rate of nonrepair in the HiLo group was 12.1% (median 12.4%).

Reasons for this wide variability were not fully delineated from the data set; however, initial comparisons between the high-volume and low-volume centers in the data set revealed similar patient demographics (Table 2). Further, comparison between the HiHi and HiLo subgroups of high volume centers revealed clinically similar (though some statistically different) patient demographics (Table 3). In addition, variability in ECMO use/availability did not seem to account for group differences. The average rates of ECMO use among the low-volume and high-volume centers were 25.9% and 27.8% ($P = 0.686$), respectively.

Preoperative risk factors found to be independently associated with mortality included birth weight, Apgar scores at 1 and

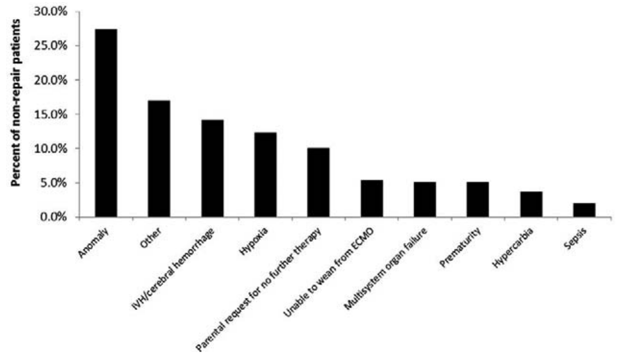

FIGURE 2. Specific reasons for nonrepair among all centers.

5 minutes, prenatal diagnosis, cardiac anomalies, and chromosomal/other anomalies. These were included in the logistic regression model: [exp(2.4010+(Birth Weight x −0.6383)+(APGAR1 x −0.1229)+(APGAR5 x −0.2895)+(Prenatal Dx x 0.9546)+(Car-(Cardiac Dx x 0.6330)+(Other Dx x 0.7667))]. The probability transformation for this log odds (logit) is logit/(1+logit). The area under the receiver operating characteristic curve for the learning set was 0.793.

Center Outcome Based on Volume and Nonrepair Rate

Mean O/E ratios for mortality among the 3 volume categories were 0.81 (99% confidence interval, CI: 0.74–0.87) for the high volume plus low rate of nonrepair group (HiLo), 0.94 (99% CI: 0.87–1.00) for the high volume plus high rate of nonrepair group (HiHi), and 1.21 (99% CI: 1.16–1.26) for the low-volume center group (Lo) (Fig. 4). Nonparametric ANOVA model P value was $P < 0.0001$. In pairwise comparisons, O/E ratios were significantly different when any 2 groups were compared ($P < 0.0001$).

Excess mortality calculations identified that the Lo group centers underperform expectations with 1.43 excess deaths per 100 patients (1.13–1.72, 95% CI), HiHi group centers have 0.68 excess deaths per 100 patients (0.32–1.04, 95% CI), and HiLo group centers overachieve expectations with 2.73 excess survivors per 100 patients (2.4–3.1, 95% CI).

DISCUSSION

Although the overall published survival rate for infants with CDH has improved over the past 25 years, many of these reports only include infants who undergo operative repair.[10] Understanding the ''true'' survival rate requires knowledge of the survival rate for both

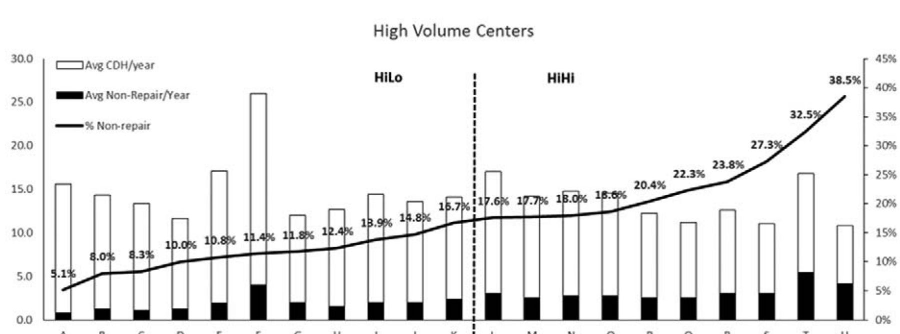

FIGURE 3. High volume centers dichotomized by rate of nonrepair. Average number of patients annually on the *y* axis and center on the *x* axis. The dashed line separates high-volume centers with a low rate of nonrepair (HiLo: 5.1%–16.7%) from those with a high rate of nonrepair (HiHi: 17.6%–38.5%).

TABLE 2. Patient Characteristics: High-volume Versus Low-volume Centers

Volume	High (>10/yr) (n = 2230)	Low (≤10/yr) (n = 1729)	P*
EGA (weeks)	37.5 ± 2.4	37.6 ± 2.4	0.372
Birthweight (g)	2.9 ± 0.6	3 ± 0.7	0.063
Apgar 1 minute	4.8 ± 2.5	4.9 ± 2.5	0.376
Apgar 5 minutes	6.7 ± 2.2	6.7 ± 2.1	0.926
Major cardiac anomaly	7.5%	8.6%	0.199
Chromosomal anomaly	5.6%	5.8%	0.786
Prenatally Diagnosed	67.8%	63.2%	0.002
Outborn	62.3%	42.1%	<0.0001

*t test or χ^2.

repaired and unrepaired infants with CDH. Since its inception in 1995, the CDHSG has captured data for more than 9500 infants diagnosed with CDH, a substantial proportion of whom never receive surgical intervention.[9] Noted in initial reports, one of the inherent features of this observational database is the inability to control for variance in practice patterns across the numerous national/international institutions.[11] Although mortality and proportion of patients repaired have a high correlation,[12] the underlying nuances of why a patient with CDH goes unrepaired are unclear and challenging to elucidate.

To understand reasons for variation in surgical patient selection, we first sought to evaluate characteristics of surgical candidates who were selected by participating centers. We found that patients selected for surgery were, not surprisingly, significantly different with generally more favorable survival characteristics including advanced gestational age, higher birth weights, and higher Apgar scores. In addition, those deemed "surgical candidates" had fewer associated major cardiac or chromosomal abnormalities. These findings align with traditional indications used to select candidates for CDH repair and reflect preference toward physiologically stable infants without associated lethal anomalies.[13,14]

There were additional important distinctions between repair and nonrepair groups. For example, patients who did not receive operative intervention were more likely to be prenatally diagnosed. This may serve as a surrogate marker for severity of illness,[15] suggesting that these patients had larger defects, with a stomach and/or liver displaced into the thorax, mediastinal/cardiac shift, and additional associated abnormalities. These factors make prenatal identification of the more severe patients more likely and/or increase the frequency of prenatal imaging and, ultimately, diagnosis. The lower proportion of outborn status in the nonrepair group may also

TABLE 3. Patient Characteristics in High-volume Centers: High Rate of Nonrepair (HiHi) Versus Low Rate of Nonrepair (HiLo)

Center group	HiHi (n = 1125)	HiLo (n = 1105)	P*
EGA (weeks)	37.3 ± 2.4	37.7 ± 2.3	<0.0001
Birthweight (g)	2.9 ± 0.6	3 ± 0.6	0.001
Apgar 1 minute	4.5 ± 2.5	5.1 ± 2.5	<0.0001
Apgar 5 minutes	6.6 ± 2.2	6.8 ± 2.1	0.035
Major cardiac anomaly	8.4%	6.6%	0.099
Chromosomal anomaly	5.2%	5.9%	0.514
Prenatally Diagnosed	68.4%	67.1%	0.501
Outborn	70.0%	54.5%	<0.0001

*t test or χ^2.

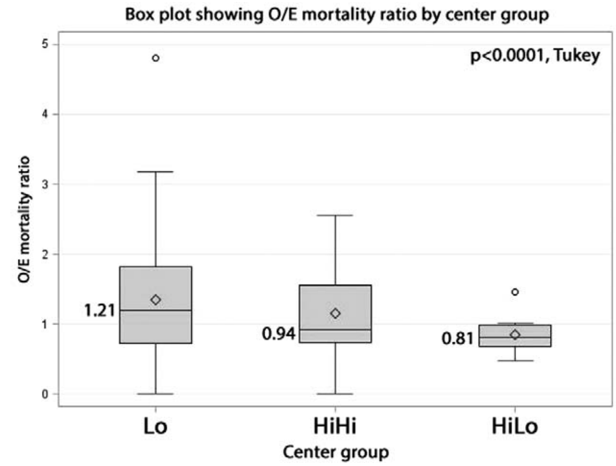

FIGURE 4. Box plot showing O/E ratio by center group. The diamond represents the mean, the line represents the median, the box represents the interquartile range, the error bars are 5th and 95th percentiles, and the circles are outliers (as defined by Tukey).

suggest a selection bias towards keeping healthier infants or, conversely, transferring more severe infants prenatally. This may also simply reflect differences in prenatal identification leading to births of undiagnosed, less-severe outborn CDH patients.

Our logistic regression model for mortality analysis consisted of preoperative variables of significance based on numerous previous publications. We elected not to use defect size, a known highly significant predictor of outcome, given the fact that this information is unknown among those patients who do not undergo repair. Further, even among those that do undergo repair, operative information is unknown to the clinician before the decision to attempt repair. We believe the variables included are not only the most significant but also include all those clinically relevant and well-defined between centers.

We identified a wide variability in nonrepair rates across centers, ranging from 0% to 66.7%. Although we have shown that as a group, healthier infants are selected for repair, the reason for the wide variability in repair rates across centers remains unknown. Most (67.7%) patients who were not repaired were considered physiologically unstable or non-survivable, a decision made by the treating institution. We examined patient-dependent factors and found no clear variation between the centers with the lowest and highest nonrepair rates, with both cohorts demonstrating similar percentages of associated cardiac and chromosomal anomalies. These findings suggest that criteria for operative intervention are likely not uniform across centers and that patient selection may depend upon institution-specific resources, experiences, and expertise.

We chose center volume categorization, in addition to the main question of nonrepair rate, given the profound variability when a few patients are managed within a given center. In centers with only a small number of patients annually, 1 or 2 patient outcomes, either positive or negative, can dramatically skew the results. This variability, combined with increased transfer and selection bias potential, make meaningful analyses challenging. Interestingly, if you dichotomize the low-volume centers the same trend (to a stronger degree) persists; however, the small patient numbers per center and complex confounding issues of patient selection render these data inappropriate for drawing absolute conclusions. Although we felt it important to include the overall low volume center data, we chose to focus on high-volume centers and then further dichotomize them. We chose 10 patients as the initial cutoff for high-volume based on previous publications[16,17] and reasonable data dichotomization. We then had a natural dichotomization in rates of nonrepair among high volume centers. Notably, patients with small defects (A & B types in the CDHSG staging system[18]) do well, irrespective of the center in which they are treated. On the other hand, patients with larger defects (C & D types) are certainly the ones most likely to go unrepaired and may account for the differences identified in this dataset/analysis.

We chose a 10-year period over which we evaluated the data for 2 reasons: (ii) permissive hypercapnia/pressure limited ventilator management and pulmonary hypertension medication use would have been a relatively standard practice at most institutions over the 10-year study period and (ii) a new version of data collection started in 2001 (several years are required to transition to the updated versions). As a result of these 2 points, we decided that data analysis for patients collected between 2004 and 2013 was the most appropriate.

CDH patients who do not undergo surgical repair comprise a small subset of the CDH population; however, their impact on overall survival may be significant. A recent study from the CAPSNet CDH registry identified a cohort of potential surgical candidates (excluding those with severe associated anomalies, those whose parents chose palliative care, and those with severe physiologic instability) who never underwent operative repair, yet were indistinguishable from the operative cohort based on EGA, birth weight, disease severity scores, ECMO requirement, and/or minor associated anomalies.[7] These findings, although seemingly inconsistent with the results presented here, are not appropriate for direct comparison, as our study did not use the same exclusion criteria. Despite this, their study highlighted significant practice pattern variability. The CAPSNet report also highlights potential opportunities for surgical intervention among a subset of patients they classified as potential surgical candidates. Although subjective and fraught with individual and institutional biases, perhaps criteria for surgical eligibility can be better defined in order to clarify these discrepancies in practice patterns.

Other reports have advocated for an aggressive management approach to CDH patients, including those with rapid and profound physiologic derangement. Kays et al[19] studied a single institution cohort of 172 patients (without severe anomalies) and identified greater than 50% survival among patients with very low Apgar scores, early need for ECMO support, profound hypercarbia (best $PCO_2>100$), and profound acidosis (pH<7.0 for >1 hour). The same group has also suggested that early recognition of the likely need for ECMO support, combined with early diaphragmatic repair prior to initiation of ECMO may confer a survival advantage.[20] Even an associated congenital malformation may not affect postnatal survival and, therefore, should be closely scrutinized as even a relative contraindication to surgical repair.[21] These reports highlight potential opportunities for survival in CDH when therapies with likely efficacy are combined with an aggressive management approach not defined by potentially arbitrary limitations.

A key contributing factor to the wide spectrum of practice patterns and outcomes across centers is the clinical heterogeneity of CDH. Attempts to risk stratify patients by markers of disease severity such as defect size, presence of associated anomalies and degree of pulmonary hypertension are critical to successful interpretation of outcomes.[9,22] Here we have confirmed higher disease severity of the nonrepair population, however the wide variability in rates of nonrepair across institutions may reflect differing treatment philosophies or resource availability. Further institution-specific data collection may offer insights into this variability in practice patterns.

There are important limitations to this study. The use of observational (though prospectively collected) data comes with the usual limitations. Further, despite consistent efforts to maintain a meticulous dataset, missing or incomplete data (though less than 1%) could affect the findings. Centers grouped by volume and rate of repair may also share unknown confounders that could account for group differences. Finally, it is impossible to identify patients who, because of prenatal information and/or after parental counseling, received only palliative care. Still, factors involved in this decision-making process (such as cardiac and chromosomal anomalies) have been accounted for in the models and the data presented here should help to inform these challenging discussions.

In conclusion, infants who undergo operative repair of CDH are significantly different than those who either are not considered surgical candidates or do not survive until operative repair. Most of these patients were deemed unstable or non-survivable; however, a small but not insignificant proportion may represent potential operative candidates and, subsequently, potential survivors. Institutional variability, even among high-volume institutions, is high. Our analyses identified more observed deaths than expected among the low volume centers and among the high-volume centers with a high rate of nonrepair, when compared with the high-volume centers with a low rate of nonrepair. For every 100 CDH patients, high volume centers with a low rate of nonrepair have at least 2.7 additional survivors beyond expectation and perform significantly better when compared with high-volume centers with high rates of nonrepair. Careful patient categorization and optimization of the opportunity for surgical intervention, along with further interrogation into practice patterns and resource availability, may identify targets for improvement and increase survival in neonatal CDH.

REFERENCES

1. Antonoff MB, Hustead VA, Groth SS, et al. Protocolized management of infants with congenital diaphragmatic hernia: effect on survival. *J Pediatr Surg*. 2011;46:39–46.
2. Tracy ET, Mears SE, Smith PB, et al. Protocolized approach to the management of congenital diaphragmatic hernia: benefits of reducing variability in care. *J Pediatr Surg*. 2010;45:1343–1348.
3. Snoek KG, Reiss IK, Greenough A, et al. Standardized postnatal management of infants with congenital diaphragmatic hernia in Europe: the CDH EURO Consortium Consensus - 2015 update. *Neonatology*. 2016;110:66–74.
4. Lazar DA, Cass DL, Rodriguez MA, et al. Impact of prenatal evaluation and protocol-based perinatal management on congenital diaphragmatic hernia outcomes. *J Pediatr Surg*. 2011;46:808–813.
5. Sola JE, Bronson SN, Cheung MC, et al. Survival disparities in newborns with congenital diaphragmatic hernia: a national perspective. *J Pediatr Surg*. 2010;45:1336–1342.
6. Aly H, Bianco-Batlles D, Mohamed MA, et al. Mortality in infants with congenital diaphragmatic hernia: a study of the United States National Database. *J Perinatol*. 2010;30:553–557.
7. Wilson MG, Beres A, Baird R, et al. Congenital diaphragmatic hernia (CDH) mortality without surgical repair? A plea to clarify surgical ineligibility. *J Pediatr Surg*. 2013;48:924–929.
8. Harting MT, Lally KP. The Congenital Diaphragmatic Hernia Study Group registry update. *Semin Fetal Neonatal Med*. 2014;19:370–375.
9. Tsao K, Lally KP. The Congenital Diaphragmatic Hernia Study Group: a voluntary international registry. *Semin Pediatr Surg*. 2008;17:90–97.
10. Congenital Diaphragmatic Hernia Study G, Lally KP, Lally PA, et al. Defect size determines survival in infants with congenital diaphragmatic hernia. *Pediatrics*. 2007;120:e651–e657.
11. Clark RH, Hardin WD Jr, Hirschl RB, et al. Current surgical management of congenital diaphragmatic hernia: a report from the Congenital Diaphragmatic Hernia Study Group. *J Pediatr Surg*. 1998;33:1004–1009.
12. Campbell BT, Herbst KW, Briden KE, et al. Inhaled nitric oxide use in neonates with congenital diaphragmatic hernia. *Pediatrics*. 2014;134: e420–e426.
13. Reiss I, Schaible T, van den Hout L, et al. Standardized postnatal management of infants with congenital diaphragmatic hernia in Europe: the CDH EURO Consortium consensus. *Neonatology*. 2010;98:354–364.
14. Skari H, Bjornland K, Haugen G, et al. Congenital diaphragmatic hernia: a meta-analysis of mortality factors. *J Pediatr Surg*. 2000;35:1187–1197.

15. Mesas Burgos C, Hammarqvist-Vejde J, Frenckner B, et al. Differences in outcomes in prenatally diagnosed congenital diaphragmatic hernia compared to postnatal detection: a single-center experience. *Fetal Diagn Ther*. 2016;39:241–247.
16. Bucher BT, Guth RM, Saito JM, et al. Impact of hospital volume on in-hospital mortality of infants undergoing repair of congenital diaphragmatic hernia. *Ann Surg*. 2010;252:635–642.
17. Hayakawa M, Ito M, Hattori T, et al. Effect of hospital volume on the mortality of congenital diaphragmatic hernia in Japan. *Pediatr Int*. 2013;55: 190–196.
18. Lally KP, Lasky RE, Lally PA, et al. Standardized reporting for congenital diaphragmatic hernia–an international consensus. *J Pediatr Surg*. 2013;48: 2408–2415.
19. Kays DW, Islam S, Perkins JM, et al. Outcomes in the physiologically most severe congenital diaphragmatic hernia (CDH) patients: Whom should we treat? *J Pediatr Surg*. 2015;50:893–897.
20. Kays DW, Talbert JL, Islam S, et al. Improved survival in left liver-up congenital diaphragmatic hernia by early repair before extracorporeal membrane oxygenation: optimization of patient selection by multivariate risk modeling. *J Am Coll Surg*. 2016;222:459–470.
21. Colvin J, Bower C, Dickinson JE, et al. Outcomes of congenital diaphragmatic hernia: a population-based study in Western Australia. *Pediatrics*. 2005;116: e356–e363.
22. Lally KP, Lally PA, Van Meurs KP, et al. Treatment evolution in high-risk congenital diaphragmatic hernia: ten years' experience with diaphragmatic agenesis. *Ann Surg*. 2006;244:505–513.

